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Studierichting: Informatica
Oefeningenreeks 1C nr. 47.4

$$z^5 = -i$$

1. Poolvorm

$$\begin{cases} r = \sqrt{0^2 + (-1)^2} = 1 \\ \theta = \tan^{-1}\left(\frac{0}{-1}\right) + \pi = -\frac{\pi}{2} \end{cases}$$
$$\Rightarrow -i = 1 \operatorname{cis}\left(-\frac{\pi}{2}\right)$$

2. Vijfde machtswortels

$$\begin{aligned} w &= \sqrt[n]{r} \operatorname{cis}\left(\frac{\theta + k2\pi}{n}\right), k \in 0, 1, 2, \dots, n-1 \\ &= \sqrt[5]{1} \operatorname{cis}\left(\frac{-\frac{\pi}{2} + k2\pi}{5}\right) \\ &= \operatorname{cis}\left(-\frac{\pi}{10} + \frac{2k\pi}{5}\right) \\ &\Rightarrow \left\{ \operatorname{cis}\left(-\frac{\pi}{10}\right), \operatorname{cis}\left(\frac{3\pi}{10}\right), \operatorname{cis}\left(\frac{7\pi}{10}\right), \operatorname{cis}\left(\frac{11\pi}{10}\right), -i \right\} \\ \text{Note: } \operatorname{cis}\left(\frac{3\pi}{2}\right) &= -i \end{aligned}$$

References